SUPPORTING ADULT GUIDANCE

Finding a side — when the unknown is in the denominator

Introduction to Trigonometry. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner extends the equation-solving method from Lesson 5 to the case where the **unknown is in the denominator** after substituting (e.g. sin 40° = 6/x). The rearrangement is now two steps — **multiply both sides by x, then divide both sides by the ratio** — so the answer is a **division**: x = known ÷ ratio. The emphasis is that this is **not a new rule** but the same equation-solving, and that the **position of x** (numerator or denominator) is what tells you whether to multiply or divide. This case arises when finding the hypotenuse (sin or cos) or the adjacent from the opposite (tan).

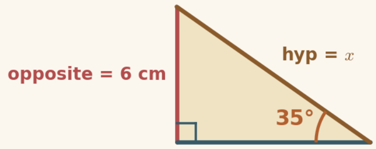
Driving Question: *What changes when the unknown is in the denominator of the ratio?*

The mathematics, in plain terms

This lesson deliberately builds on the last one to make a point about **understanding versus rules**. In Lesson 5 the unknown was in the numerator of the ratio, so a single multiplication solved it. A learner who memorised 'multiply' is now stuck, because today the unknown often lands **in the denominator**. A learner who **understood the equation** simply looks at where x is and carries on. Take finding the hypotenuse when the opposite is 6 and the angle is 40°: **(1)** sin 40° = opposite/hypotenuse; **(2)** substitute with the unknown as x — sin 40° = 6/x, and notice x **is in the denominator**; **(3)** multiply both sides by x to get sin 40° × x = 6, then divide both sides by sin 40°, giving x = 6 ÷ sin 40°; **(4)** evaluate, x = 6 ÷ 0.643 = 9.3 cm. The key teaching move is to **read the operation from the equation**: if x is in the numerator you multiply; if x is in the denominator you multiply by x and then divide, ending in a division. Insisting on all four lines — especially writing the substituted equation so the position of x is visible — is what makes the choice obvious. This case occurs whenever the **hypotenuse** is unknown (sine or cosine) or the **adjacent is found from the opposite** using the tangent. The common error is to carry over 'multiply' from the last lesson without checking where x is; the antidote is always to write line 2 and look. Getting comfortable moving between the two cases is exactly the fluency needed before finding *angles* next.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

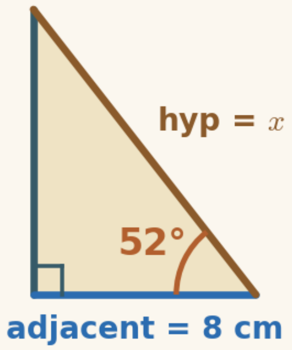


Example 1 — the unknown in the denominator (sine)

Here the opposite side is known (6 cm) and the hypotenuse is the unknown. When you write the ratio, x ends up in the denominator — so you divide.

1. write the ratio	$\sin 35° = \dfrac{\text{opposite}}{\text{hypotenuse}}$
2. put in the numbers	$\sin 35° = \dfrac{6}{x}$
3. rearrange (x is in the denominator, so divide)	$x = 6 \div \sin 35°$
4. work it out	$x = 6 \div 0.574$

x = 10.46 cm



Example 2 — the unknown in the denominator (cosine)

This time the adjacent is known (8 cm) and the hypotenuse is the unknown. Again x is in the denominator, so you divide.

1. write the ratio	$\cos 52° = \dfrac{\text{adjacent}}{\text{hypotenuse}}$
2. put in the numbers	$\cos 52° = \dfrac{8}{x}$
3. rearrange (x is in the denominator, so divide)	$x = 8 \div \cos 52°$
4. work it out	$x = 8 \div 0.616$

x = 12.99 cm

Read the operation from the equation

The heart of this lesson is **transfer**: recognising a situation that looks new as the same idea seen from a different angle. It is worth being explicit with the learner that there is **no second rule** — only the same four lines and a look at where x sits. Two supports matter. First, always **write line 2** (the substituted equation) so the position of x is visible; the whole decision — multiply or divide — is read off from there. Second, make the **two-step rearrangement** explicit at least a few times: multiply both sides by x (which lifts x out of the denominator), then divide both sides by the ratio. Once a learner has done that a handful of times, they can go straight to x = known ÷ ratio with understanding, not memory. Mixing numerator and denominator problems (as the Reflection cornerstone does) is deliberate: it forces the learner to check the equation every time rather than pattern-matching on the previous question. That habit — reason from the equation, do not guess from the last example — is the real learning.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Numerator or denominator?: choose the ratio, then reveal the two-step rearrangement (× by x, ÷ by the ratio) one line at a time.	~10 min
Movement	Side Finder (rearrange): choose the ratio, type the division rearrangement, then evaluate on the calculator.	~12 min
Reflection	Multiply or divide?: read where x sits — in the numerator → multiply; in the denominator → divide. Mixed cases, every time.	~12 min
Creativity	Reach the unmeasurable (the long way): find a sloping length (a hypotenuse) — a ladder, a wire, a ramp.	~6 min
Rest	A pause after seeing 'being in the denominator means divide' fall out of the equation, not a new rule.	~3 min
Nutrition	Mode B (intellectual): what the learning feeds — transfer, and understanding over memorised cases.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Carrying over 'multiply' from last lesson without checking

Repair: "Write line 2. Where is x — in the numerator or the denominator? In the denominator means you'll divide."

Not writing the substituted equation, so the position of x is hidden

Repair: "Always write ratio = ?/? with x in place. The equation shows you what to do."

Doing only one step of the rearrangement

Repair: "x is in the denominator, so first multiply both sides by x, then divide both sides by the ratio."

Choosing the wrong ratio

Repair: "Which two sides are in the problem — the known one and the one you want? Use the ratio with exactly those two."

Calculator in radians, not degrees

Repair: "Check the mode says DEG — sin 30° should be exactly 0.5."

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: finding a side when the unknown is in the numerator (Lesson 5), the three ratios and SOH · CAH · TOA (Lesson 4), and rearranging an equation where the unknown is in the denominator (multiply both sides by x, then divide). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the learner reading the operation from the equation, not from memory:

- “Write line 2. Is x in the numerator or in the denominator?”
- “In the denominator — so multiply both sides by x. What does that give?”
- “Now divide both sides by the ratio. What is x equal to?”
- “A ladder up a wall — the ladder is the hypotenuse, so x is in the denominator. Divide.”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“How do I know whether to multiply or divide?”
Write the substituted equation and look at x. If x is in the numerator, multiply both sides by the known number. If x is in the denominator, multiply both sides by x and then divide by the ratio — the answer is a division. You read it from the equation, not from a rule.

“When does the unknown end up in the denominator?”
Whenever you are finding the hypotenuse (using sine or cosine), or finding the adjacent from the opposite (using the tangent). In those cases the known side is in the numerator and x is in the denominator.

“Is this a different method from last lesson?”
No — it is exactly the same four lines: ratio, substitute, rearrange, evaluate. The only difference is that the rearrangement takes an extra step because x is in the denominator, so the answer is a division.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.

Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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